Supermicro 2U Hyper A+ Server (AS -2125HS-TNR)

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SanDisk
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MLPerf Storage v2.0
Single Host Local Storage
CLOSED

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1. Hardware

The hardware is composed of the client and server being on the same single host. No switch is involved.

1.1 Client

The client is the same as the Server

1.2 Server

The server is a commercially available:

Supermicro 2U Hyper A+ Server (AS -2125HS-TNR)

RAM: 256GB : 2 Socket AMD Processors (EPYC-9354-32 core)

Direct Attached (PCI-e Gen5x4): Sandisk DC SN861 NVMe

Power:

1600W Redundant Power

The configuration has NVMe drive with 15.36TB capacity. The Drive is connected via pci-e gen 5x4 lanes. The server communicates with the client through localhost as it is the same system.

1.3 Switch

There is no switch involved, as it involves only one rack server utilizing localhost to connect.

2. Software

2.1 Client

The clients distribution is the same as section 2.2 described below, as it is a single rack server.

2.2 Server

The server's distribution is Ubuntu 24.04.1 LTS with kernel 6.8.0-60-generic. The server is using an XFS file system on a single drive to host the training and checkpointing files.

2.3 Switch

There is no switch involved, as it involves only one rack server utilizing localhost to connect.

3. Settings

3.1 Client

Refer to section 3.2 as it is the same rack server.

3.2 Server

No tuning was applied on the server or the OS.

3.3 Switch

No tuning was applied, as no switch was involved, and everything was done via localhost.

4. Power Requirements

Power Requirements Table

System component Nameplate rated power Design power:

1600W Redundant Power

SanDisk SN861 NVMe 25 watts: U2 FF

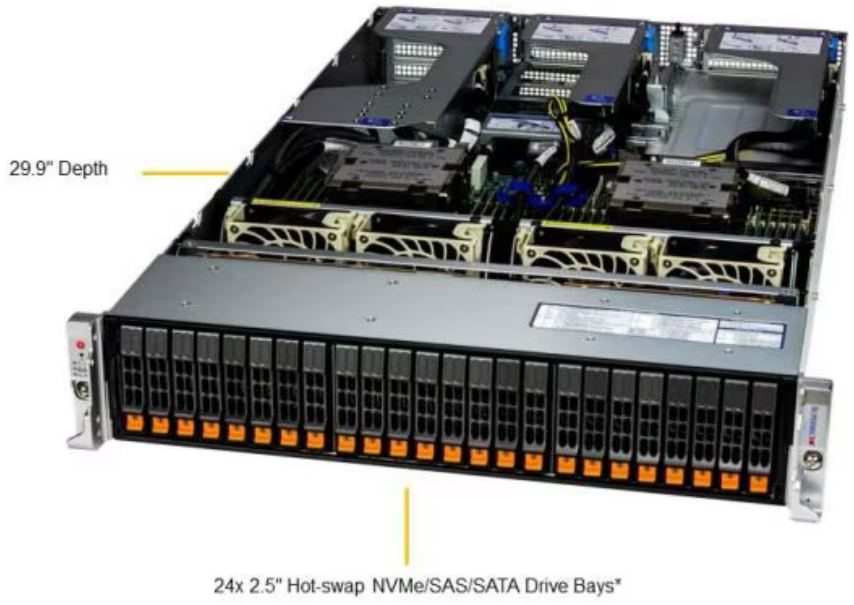
5. System Topology

SYSTEM Configuration : For testing unet3d training MLPerf Storage 2.0

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Direct Attached (PCI-e Gen5x4): Sandisk DC SN861 NVMe U2 - FF



*NVMe/SAS/SATA support requires additional parts in optional parts list

*Actual product may differ from image due to various configurations